



$(\mathcal{M}_u^*, \mathcal{M}_\ell^*)$



$\mathcal{M}'_\ell$

Novel Pose θ'

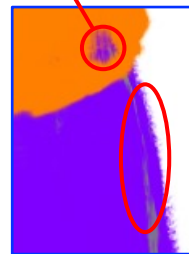


$(\mathcal{M}_u^*, \mathcal{M}'_\ell)$

Penetration
detection



$\hat{I}'$



$\hat{S}'$

Penetration-aware
Rendering



$(\mathcal{M}_u^*, \mathcal{M}'_\ell)$